

AI Implications: Multi-Agent Protocols and Emergent Alignment

Multi-Agent Communication: Compositional AI Protocols

The categorical framework developed in the preceding sections—case categories, functorial semantics, enriched structure, and diagrammatic reasoning—addresses a core challenge in contemporary artificial intelligence: how to endow autonomous agents with structured, compositional communication.

Current multi-agent AI systems communicate through increasingly standardized protocols. Google's Agent-to-Agent (A2A) protocol [a2a2025], introduced in 2025, provides a standard way for agents to communicate regardless of underlying framework through HTTP/JSON-RPC message passing with capability discovery. The Model Context Protocol (MCP) [mcp2024] standardizes how AI agents access external tools and data sources. The Agent Communication Protocol (ACP) [acp2025] handles standardizing messaging formats across various users, including agents, applications, and humans. The Agent Network Protocol (ANP) [anp2025] introduces a three-layer architecture supporting trusted agent interaction in distributed systems. These engineering advances solve critical interoperability problems but

Case Roles as Functional Typing for Agent Systems

The eight-case framework of CEREBRUM [friedman2024cerebrum] maps directly onto the functional roles that components play in multi-agent architectures:

Case Role	Agent System Analogue	Protocol Function
NOM (Agent)	Active requester	Initiator of action policies
ACC (Patient)	Passive responder	Target of directed operations
GEN (Source)	Context provider	Supplier of prior information
DAT (Recipient)	Designated receiver	Endpoint for information transfer
INS (Instrument)	Tool / API	Means of state transformation
LOC (Context)	Environment state	Boundary conditions and Markov blanket

Categorical Deep Learning: An Algebraic Theory of Architectures

Gavranović [-@gavranovic2024thesis] develops “a novel mathematical foundation for deep learning based on the language of category theory,” showing that neural network architectures—feedforward, convolutional, recurrent, attention—can all be subsumed under a single categorical framework using parameterized optics and lenses. This “Categorical Deep Learning” approach [@shiebler2024categorical] demonstrates that categorical structure is “an algebraic theory of all architectures,” unifying design patterns that appear unrelated when viewed through conventional linear algebra.

For case theory in AI, this result is significant: it means the same categorical language used to describe linguistic case assignment also describes the internal computation of the neural networks that process language. A transformer’s attention mechanism, viewed through Gavranović’s lens, performs a kind of *categorical case assignment*—each attention head selects which input tokens play

String Diagrams as a Foundation for Interpretable AI

The lambeq library [lorenz2023lambeq] demonstrates that string diagrams provide a practical interface between linguistic structure and machine learning. As an “efficient high-level Python library for Quantum NLP,” lambeq translates sentences into string diagrams, converts diagrams into parameterized circuits (quantum or classical), and trains parameters end-to-end on NLP tasks. The diagrammatic representation serves dual purposes: it is both the mathematical specification of the model *and* a human-readable explanation of what the model computes.

This interpretability property is crucial for AI safety and alignment. Where transformer architectures produce opaque attention patterns, a DisCoCat model deployed via string diagrams produces derivation trees with explicit compositional semantics—every box and wire has a linguistic interpretation. Extending this to agent communication, a categorical protocol would produce not just working message exchanges but *interpretable* interaction diagrams where each step’s relational role is transparent.

DisCoCirc and Multi-Turn Agent Dialogue Protocols

The DisCoCirc framework [defelice2022discocirc] extends compositional semantics from single sentences to multi-sentence discourse by introducing *state wires* that persist across sentence boundaries. This architecture maps directly onto multi-turn agent dialogues:

- ▶ **Entity wires** correspond to persistent agent identities across communication rounds
- ▶ **State updates** correspond to belief revisions triggered by incoming messages
- ▶ **Coreference resolution** corresponds to entity tracking across protocol sessions
- ▶ **Discourse coherence** corresponds to protocol correctness constraints

In a multi-agent system, a DisCoCirc-style protocol would track the evolving states of all participating agents as persistent wires in a circuit diagram, with each message exchange represented as a box that transforms the relevant wires. Protocol correctness

Compositional Game Theory and Categorical Multi-Agent Equilibria

The “parametrised optics” framework developed within categorical cybernetics “provides a general-purpose foundation for the study of controlled processes” applicable to compositional game theory as a multi-agent framework. In this setting, agents are modeled as lenses (or optics) that observe the environment through one channel and act through another, with the overall system behavior emerging from the composition of individual agent behaviors.

This connects to our enriched case framework (??): the precision weights on morphisms correspond to the utility parameters of game-theoretic agents, and the composition inequality $\mathcal{C}(A, C) \geq \mathcal{C}(A, B) \cdot \mathcal{C}(B, C)$ corresponds to the sub-optimality of indirect communication chains. Equilibria in the multi-agent game correspond to fixed points of the enriched functor, where no agent can improve its utility by changing its case-role assignment.

Double Categorical Systems Theory for Explainable Autonomous AI

Recent work on Double Categorical Systems Theory (DCST) aims to “utilise Double Categorical Systems Theory as a mathematical framework to facilitate collaboration in co-designing an explainable and auditable model of an autonomous AI system’s deployment environment.” DCST extends ordinary category theory with a second dimension of morphisms (2-morphisms), allowing simultaneous modeling of:

1. **Horizontal composition:** Sequential chaining of agent interactions (morphism composition)
2. **Vertical composition:** Hierarchical nesting of subsystems within larger systems (2-morphism composition)

For case theory, the double-categorical extension allows us to model both the *within-sentence* case structure (horizontal) and the *discourse-level* case structure (vertical) within a single algebraic framework—precisely the unification that DisCoCirc achieves pragmatically.

Toward Categorical Communication Standards for AI Agents

The synthesis of these developments suggests a research program: developing **categorical communication protocols** that combine the engineering robustness of existing standards (A2A, MCP, ACP) with the compositional semantics of categorical linguistics. Such a protocol would:

1. **Type-check interactions:** Every message exchange would be relationally typed by case roles, preventing structural communication errors at the protocol level
2. **Compose transparently:** Multi-step interactions would compose algebraically, with diagrammatic representations providing interpretable audit trails
3. **Transfer across implementations:** Topos-theoretic bridges (??) would ensure that a protocol verified in one categorical formalization automatically satisfies correctness conditions in any Morita-equivalent formulation
4. **Scale via enrichment:** Distributional proximity measures (??) would enable cross-formal verification and constraint